Calendar, exchange-rate and news effects in Swiss retail card spending

Course	Data Collection, Integration and Preprocessing (CIP)	**Semester**	HS2026
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Git Repository	CIP_HS2026_203		

1. Introduction

Nowadays card payment contributes to the vast majority of everyday household spending. This project aims to use daily card-payment data from **Monitoring Consumption Switzerland** to explain how retail spending behaviour in Switzerland's seven Grossregionen is affected by the calendar, the EUR/CHF exchange rate and how major public events affect spending in the Zürich canton. Our time window will cover the full years **2019 to 2025** and up to seven spending categories. It is important to note that the target variable is the scaled spending value; and for that reason, all results are reported as percentage changes. The project will have an explanatory focus rather than predictive: the scope is to quantify and understand spending behaviour, not to forecast it.

2. Research Questions

Q1: *How do major events (# of participants) in Canton Zürich affect daily POS spending across different merchant categories compared with similar non-event days?*

Q2: *How much of the daily variation in spending by category is explained by calendar effects (weekday, month, public holidays)? Specifically, how does a regional public holiday change spending by category in regions where it applies, compared with regions where the same date is a normal working day?*

Q3: *Is spending in border regions more strongly affected by a decrease in EUR/CHF exchange rate than spending in non-border regions, and does this effect differ between border regions?*

3. Data Sources and Acquisition

Source	Content	Method of Acquisition
Monitoring Consumption Switzerland	Daily scaled card spending by Grossregion and category, 2019-2025	Download
SNB data portal	Daily EUR/CHF exchange rate, 2019-2025	API requests
feiertagskalender.ch	Public holidays per canton, 2019-2025	Scraping (Selenium + BeautifulSoup)
swissinfo.ch or local cantonal alternatives	Title, date, # of participants, in Zurich, 2019-2025	Scraping (Selenium + BeautifulSoup)
Federal Statistical Office (BFS)	Permanent resident population per canton	Download

Before scraping, the robots.txt of both websites will be checked; requests will be rate-limited, and the raw HTML/JSON is stored once so that scraping does not need to be repeated.

4. Data Cleaning and Transformation

- **Missing data:** A comprehensive data quality assessment will be initially performed to evaluate potential gaps in the data that could invalidate our research questions, including checking every series for gaps; filling missing days (e.g. due to holidays) by carrying the last available rate forward (LOCF); decide what to set as 0 or NAs.
- **Datatypes:** Parse dates to datetime, regions and categories to categorical, spending and rates to float.
- **Ranges and outliers:** drop region category 9999 (unidentified); check for logical outliers (e.g. negative spending); flag COVID periods instead of deleting them; detect outliers (e.g. extreme spending using IQR).
- **Merging:** convert canton holidays to Grossregion level using population weights retrieved from BFS dataset, giving the share of residents who have the day off (0 to 1); remove duplicate news articles returned by several searches; merge all sources on date and region into one panel.
- **Add columns:** holiday share, weekday/month/year variables, border-region indicator, spending and the event size.

5. Planned Analysis

Q1: compare spending on event days with comparable non-event days, controlling for weekday, seasonality and public holidays. Analyse the percentage difference in scaled spending across selected merchant categories, particularly Food & Beverage, Accommodation and Transport.

Q2: regression of daily spending on the regional holiday share, controlling for region, category, weekday, month and year; a holiday × category interaction makes it possible to interpret the effect per category.

Q3: regression of daily spending on the 30-day average EUR/CHF rate with an EUR/CHF × border-region interaction, controlling for holidays, calendar effects and the COVID flag.

Results are visualised with time-series plots, heatmaps of holiday effects by region and category, and coefficient plots (matplotlib/seaborn).

6. Risks and Mitigation

Risk	Mitigation / backup plan
News website changes its layout, restricts access or disallows paths in robots.txt	Scrape once with delays and keep the raw HTML; backup: another Swiss news source (e.g. SRF News archive or RSS feeds) or ultimately switching to API retrieval
Holiday website changes	Scrape 2019-2025 once and store the raw HTML and the resulting CSV, so later changes cannot affect the project Backup: retrieval using OpenHolidays API
COVID-19 causes structural breaks in spending (2020–2021)	Flag periods and include them as controls; robustness check excluding 2020-2021
Scaled spending values, no CHF levels	Interpret only relative effects (percentage changes) within categories

7. Use of Generative AI

Generative AI use will be limited to debug code and to polish the language of the text. All ideas, analyses and conclusions are the result of collaborative teamwork. Any correction suggested by Generative AI will be strictly evaluated by all three members altogether and discarded if needed.